Macroeconomics I
Problem Set 2

1. **(Neoclassical Growth Model with Exchanges in Period 0)**. Consider the neoclassical growth model. There is a representative family with unit mass. There is no population growth. The utility function is given by:

$$U(c_{t t=0}^{\infty}) = \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma} - 1}{1-\sigma},$$

with k_0 given. The final good that can be consumed or invested is produced using the following production function: $y_t = k_t^{\alpha} n_t^{1-\alpha}$. The accumulation of capital is given by the following motion law:

$$k_{t+1} = k_t(1-\delta) + i_t.$$

   (a) Describe the household and the firm problem in an Arrow-Debreu market structure (exchanges in period 0).

   (b) Characterize the equilibrium: solve the problems and interpret the solutions. Write down the system of difference equations that characterize the solution of this economy (including initial and terminal conditions).

   (c) Define a competitive equilibrium for this economy.

2. **(An Economy with Two Sectors)**. Consider the following economy. The representative household has utility function:

$$U = \sum_{t=0}^{\infty} \frac{1}{(1+\rho)^t} u(c_t), \quad \rho > 0.$$

Where $u(.)$ is strictly increasing, strictly concave, differentiable, bounded with $\lim_{c \to 0} u'(c) = \infty$. The household is endowed with one unit of time.

The economy is composed of two sectors, one for the production of consumption goods and another for the production of investment goods. The technology of the consumption goods sector is

$$c_t = A k_{c_t}^{\theta} n_{c_t}^{1-\theta}, \quad 0 < \theta < 1.$$

The technology of the investment goods sector is

$$x_t = B k_{x_t}^{\phi} n_{x_t}^{1-\phi}, \quad 0 < \phi < 1.$$

The capital accumulation technology is

$$k_{t+1} = k_t(1-\delta) + x_t, \quad 0 < \delta < 1,$$

and the feasibility constraints of the economy are:

$$n_{c_t} + n_{x_t} \leq 1 \quad \text{and} \quad k_{c_t} + k_{x_t} \leq k_t.$$

Suppose the household owns the capital stock and sells the services of capital and labor, whose prices in terms of the consumption good in period t are given by r_t and w_t, respectively. The price of the investment good in terms of the consumption good in period t is q_t. After receiving factor income, households decide how much to consume and invest. Households are born with initial capital k_0.

(a) Carefully define the problem of the household and the problem of the firms (in both sectors) in a sequential market environment.

(b) Define a competitive equilibrium for this economy.

(c) Define the intertemporal optimization problem solved by a benevolent central planner for this economy.

(d) Obtain and interpret the necessary first-order conditions for this problem.

(e) Specify a set of equations that characterize the equilibrium quantities in the steady state. (Hint: this system of equations involves the steady-state quantities c, x, k_c, k_x, k, n_c and n_x, and the steady-state prices r, q, and w.)

(f) Specify an algorithm to compute the quantities c, x, k_c, k_x, k, n_c and n_x, and the prices r, q, and w in the steady state. It is not necessary to solve the problem algebraically.

3. **(Baby Boom Dynamics).** Consider the standard Ramsey-Cass-Koopmans model with population growth (but no technological growth). The preference of the representative household is given by:

$$\max_{c_t \geq 0} \int_0^\infty e^{-(\rho-n)t} \frac{c_t^{1-\sigma}}{1-\sigma} dt, \tag{1}$$

where c_t is per capita consumption. The resource constraint (per capita) of the economy is given by:

$$\begin{aligned} f(k_t) &= c_t + i_t, \\ \dot{k}_t &= i_t - (n+\delta)k_t. \end{aligned}$$

Suppose $\rho > n$ for all changes in n in the questions.

(a) Characterize the planner's solution for this economy.

(b) Derive the equations characterizing the steady state and draw the phase diagram.

(c) Consider a permanent and unannounced increase in the population growth rate of the economy. Suppose the economy is in steady state at the time of the increase and show the dynamics in the phase diagram.

(d) Now consider a *temporary* and unannounced increase in the growth rate at t_0. That is, $n_t = n' > n$ for $t_0 \leq t \leq T$, but $n_t = n$ for $t > T$. Suppose the economy is in steady state at t_0 and show the dynamics of consumption and investment in the phase diagram and over time (*Hint*: consumption jumps only when the change in n is announced).

(e) Now consider a permanent increase in the growth rate at T, *but announced at* $t_0 < T$. Suppose the economy is in steady state at t_0. Show the dynamics of consumption and investment in the phase diagram and over time.